R PROGRAMMING

EXPERIMENT-1

ADDITION:

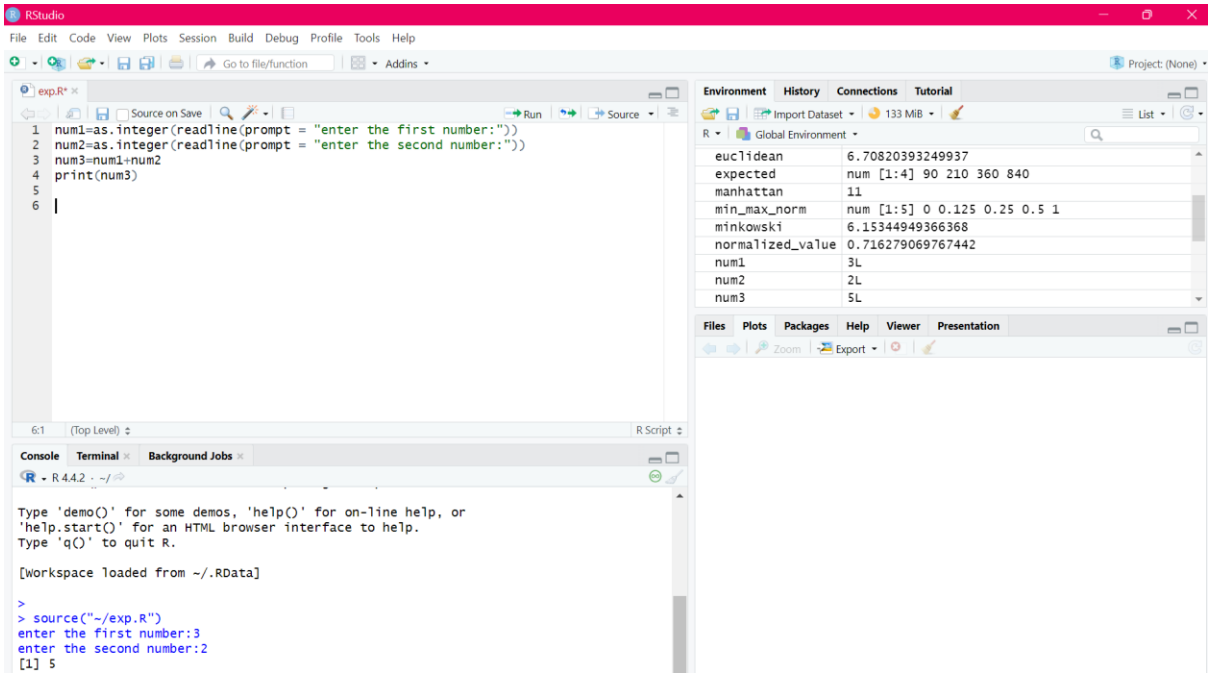
AIM:

To prove the program for addition using R-tool.

PROGRAM:

```
num1=as.integer(readline(prompt = "enter the first number:"))
num2=as.integer(readline(prompt = "enter the second number:"))
num3=num1+num2
print(num3)
```

OUTPUT:



RESULT:

Thus the basic program addition is executed successfully.

EXPERIMENT-2

SUBTRACTION:

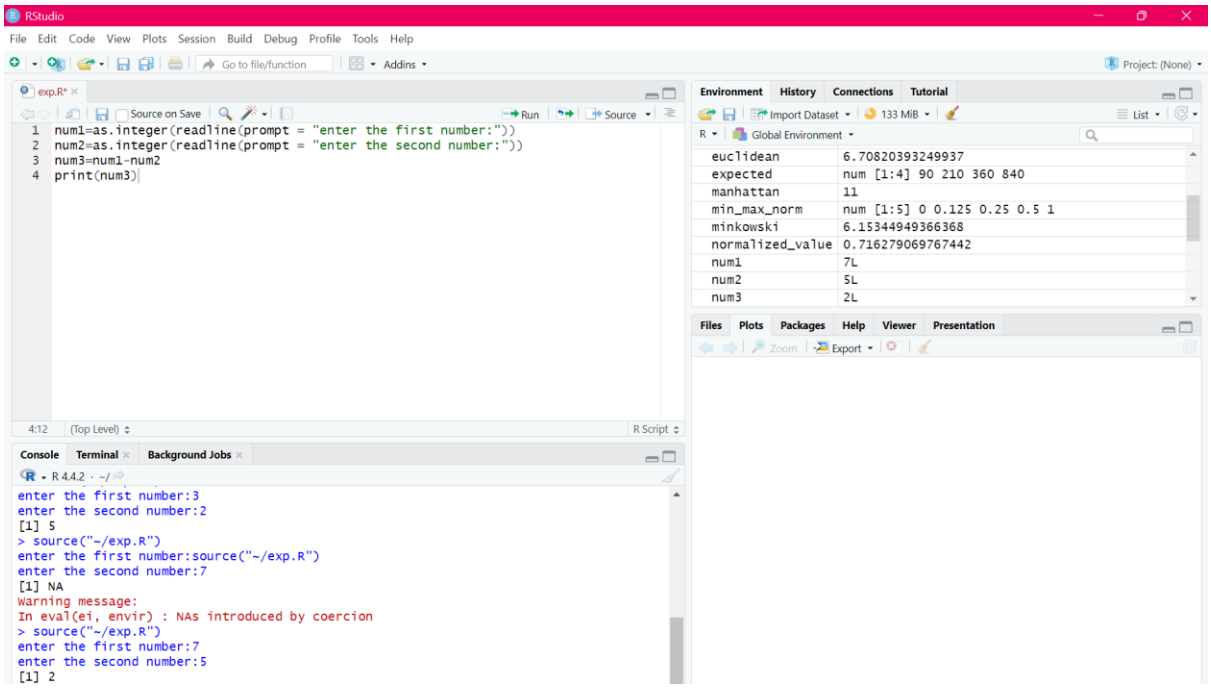
AIM:

To prove the program for subtraction using R-tool.

PROGRAM:

```
num1=as.integer(readline(prompt = "enter the first number:"))
num2=as.integer(readline(prompt = "enter the second number:"))
num3=num1-num2
print(num3)
```

OUTPUT:



RESULT:

Thus, the basic program subtraction is executed successfully.

EXPERIMENT-3

MULTIPLICATION:

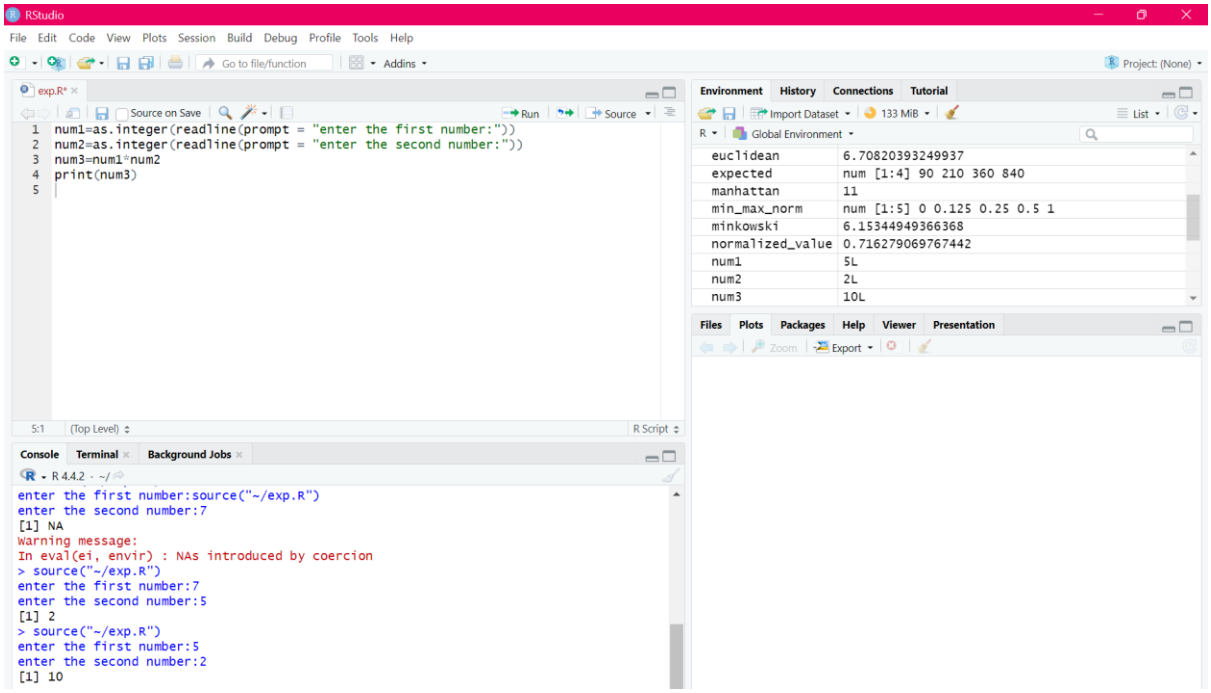
AIM:

To prove the program for multiplication using R-tool.

PROGRAM:

```
num1=as.integer(readline(prompt = "enter the first number:"))
num2=as.integer(readline(prompt = "enter the second number:"))
num3=num1*num2
print(num3)
```

OUTPUT:



RESULT:

Thus, the basic program multiplication is executed successfully.

EXPERIMENT-4

DIVISION:

AIM:

To prove the program for division using R-tool.

PROGRAM:

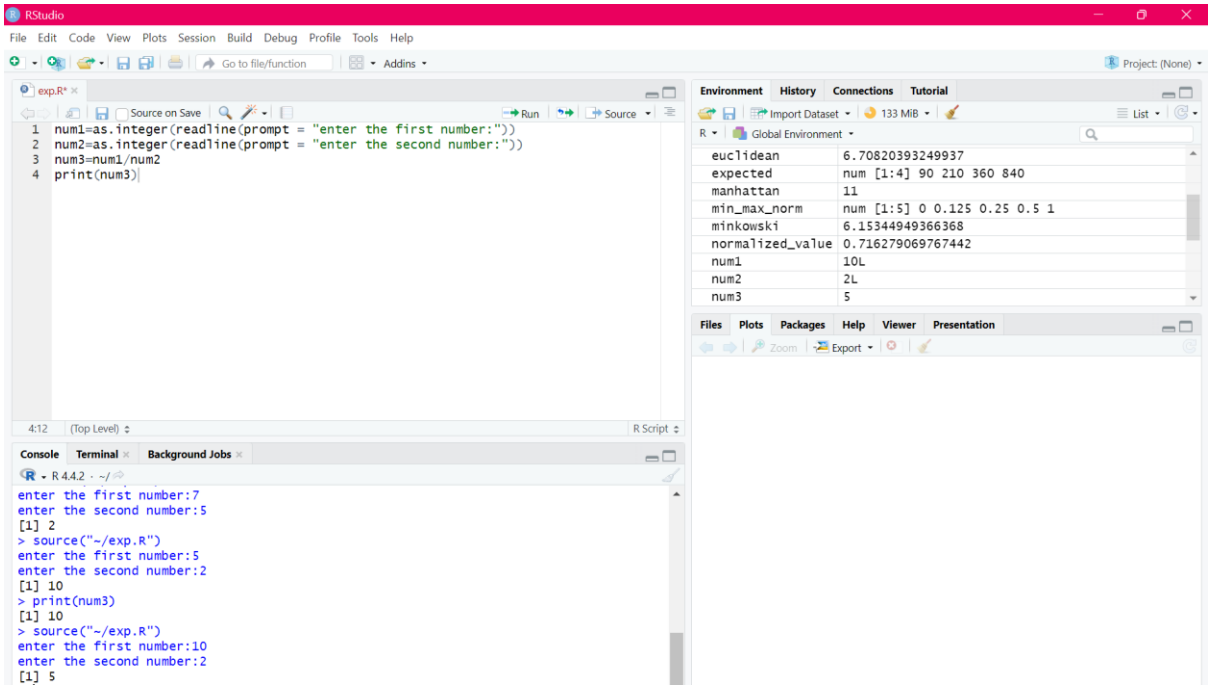
num1=as.integer(readline(prompt = "enter the first number:"))

num2=as.integer(readline(prompt = "enter the second number:"))

num3=num1/num2

print(num3)

OUTPUT:



RESULT:

Thus, the basic program division was executed successfully.

EXPERIMENT-5

ODD OR EVEN:

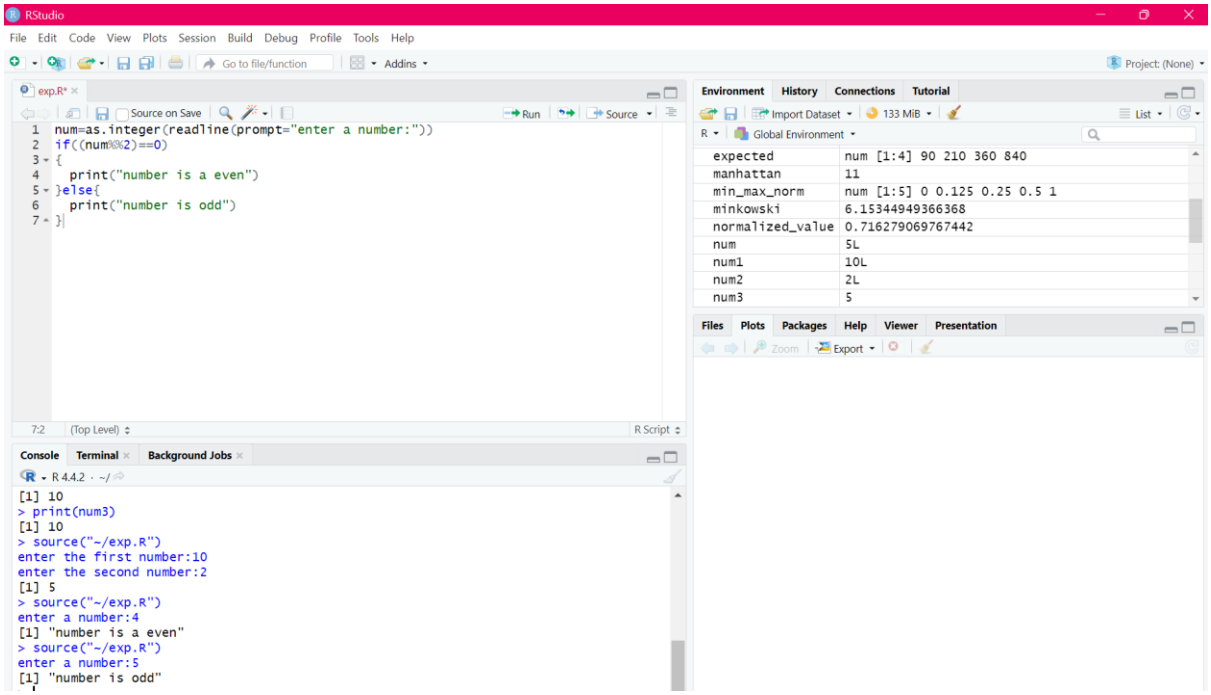
AIM:

To write the program for odd or even using R-tool.

PROGRAM:

```
num=as.integer(readline(prompt="enter a number:"))
if((num%%2)==0)
{
  print("number is a even")
}else{
  print("number is odd")
}
```

OUTPUT:



RESULT:

Thus, the basic program odd or even was executed successfully.

EXPERIMENT-6

MEAN,MEDIAN,MODE:

AIM:

To write the program for mean,median,mode.

PROGRAM:

MEAN

```
names<-c("siri","mahi","chiru")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
mean(df $age)
write.csv(df,"datafr.csv")
```

MEDIAN

```
names<-c("siri","mahi","chiru")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
```

median(df $age)

write.csv(df,"datafr.csv")

MODE

names<-c("siri","mahi","chiru")

age<-c(23,24,25)

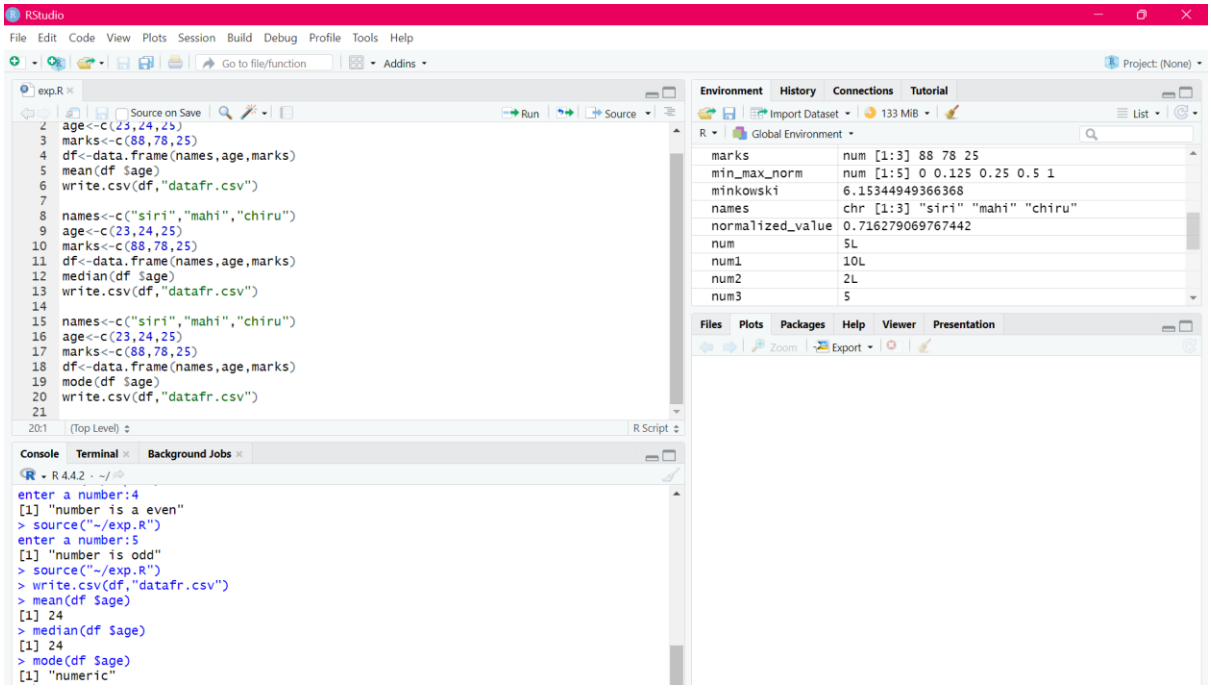
marks<-c(88,78,25)

df<-data.frame(names,age,marks)

mode(df $age)

write.csv(df,"datafr.csv")

OUTPUT:



RESULT:

Thus the central tendency and measure of dispersion is executed successfully.

EXPERIMENT-7

SUMMARY:

AIM:

To write the program for summary using R-tool.

PROGRAM:

names<-c("siri","mahi","chiru")

age<-c(23,24,25)

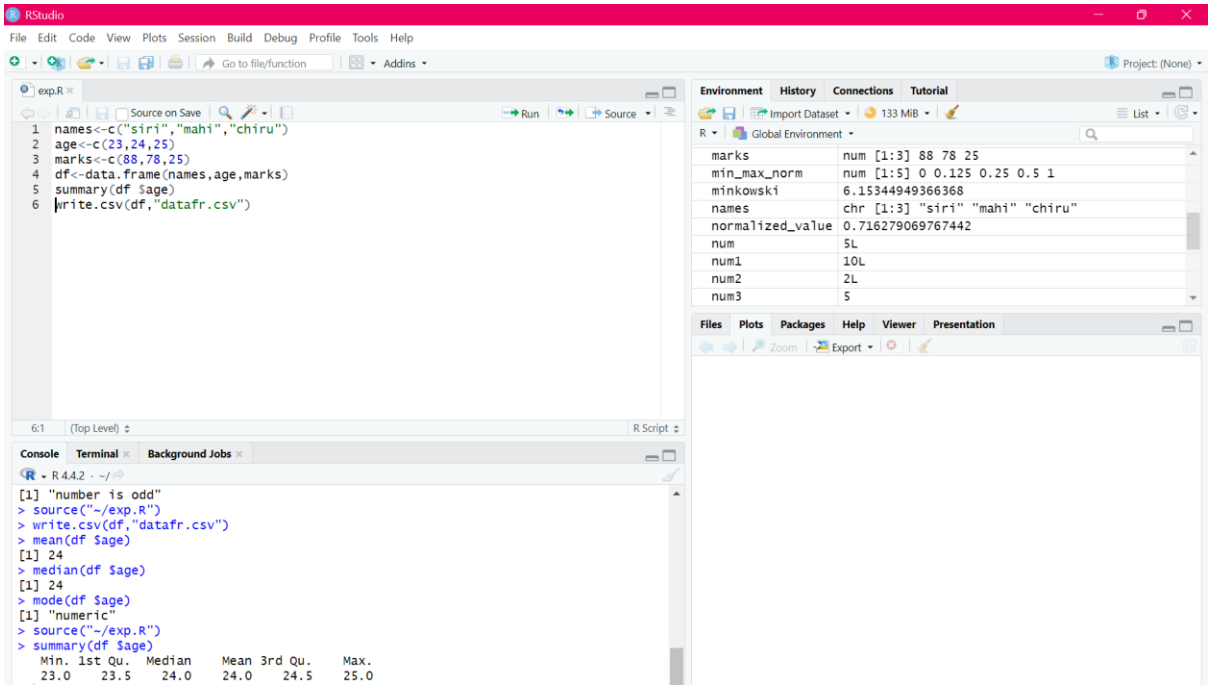
marks<-c(88,78,25)

df<-data.frame(names,age,marks)

summary(df $age)

write.csv(df,"datafr.csv")

OUTPUT:



RESULT:

Thus, the central tendancy and measure of dispersion is executed successfully.

EXPERIMENT-8

GREATER AMONG THREE NUMBERS:

AIM:

To write the program for the greatest among three numbers.

PROGRAM:

```
x <- as.integer(readline(prompt = "Enter first number :"))

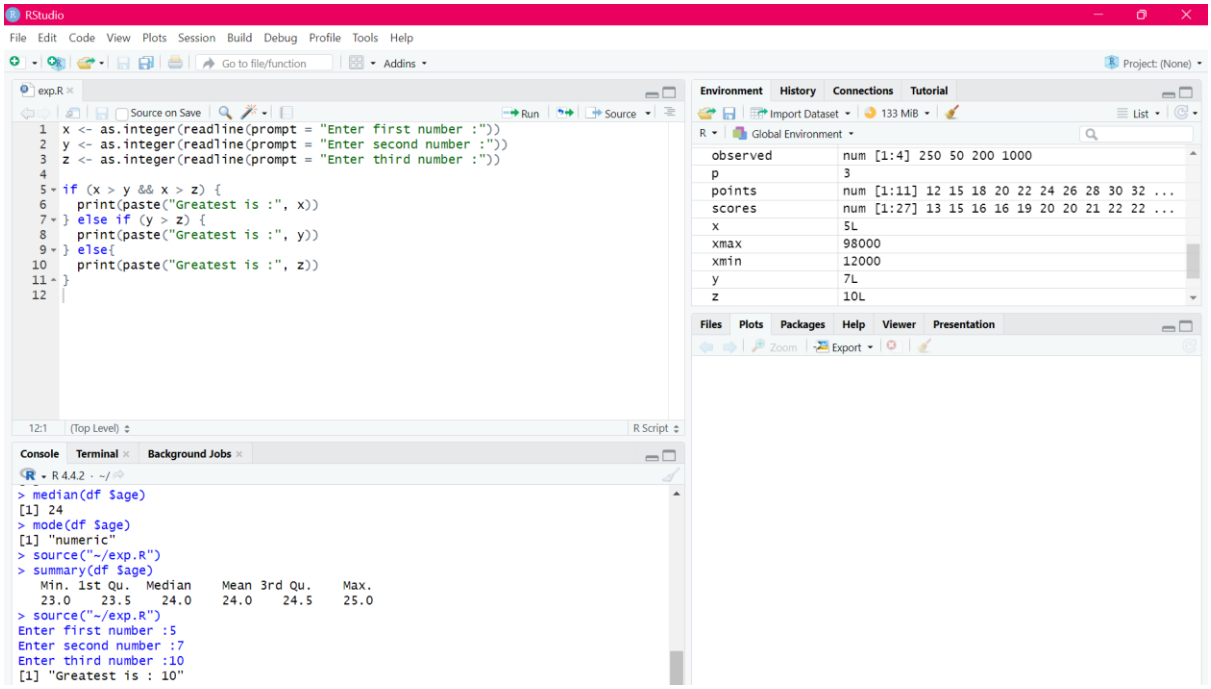
y <- as.integer(readline(prompt = "Enter second number :"))

z <- as.integer(readline(prompt = "Enter third number :"))


if (x > y && x > z) {
  print(paste("Greatest is :", x))
} else if (y > z) {
  print(paste("Greatest is :", y))
```

```
} else{
  print(paste("Greatest is :", z))
}
```

OUTPUT:



RESULT:

Thus, the greatest among the three numbers was executed successfully.

EXPERIMENT-9

IQR:

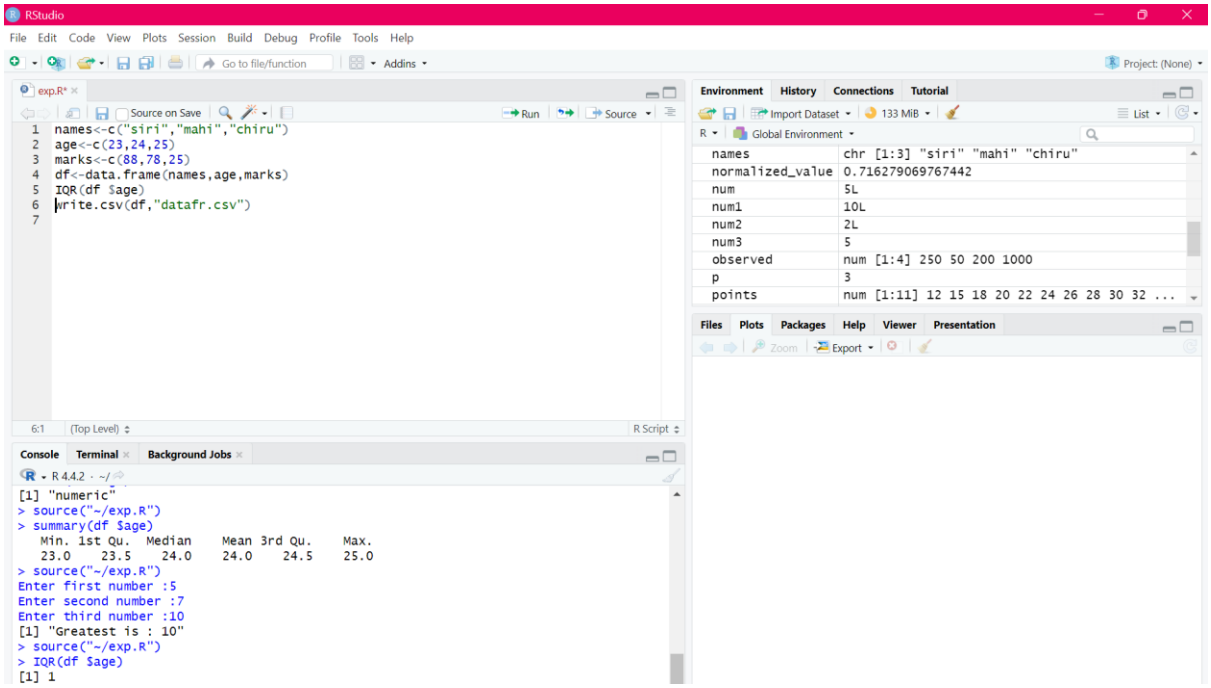
AIM:

To write the program for central tendency and data dispersion measures using R tool.

PROGRAM:

```
names<-c("siri","mahi","chiru")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
IQR(df $age)
write.csv(df,"datafr.csv")
```

OUTPUT:



RESULT:

Thus, the program for central tendency and data dispersion measures was executed successfully.

EXPERIMENT-10

QUANTILE:

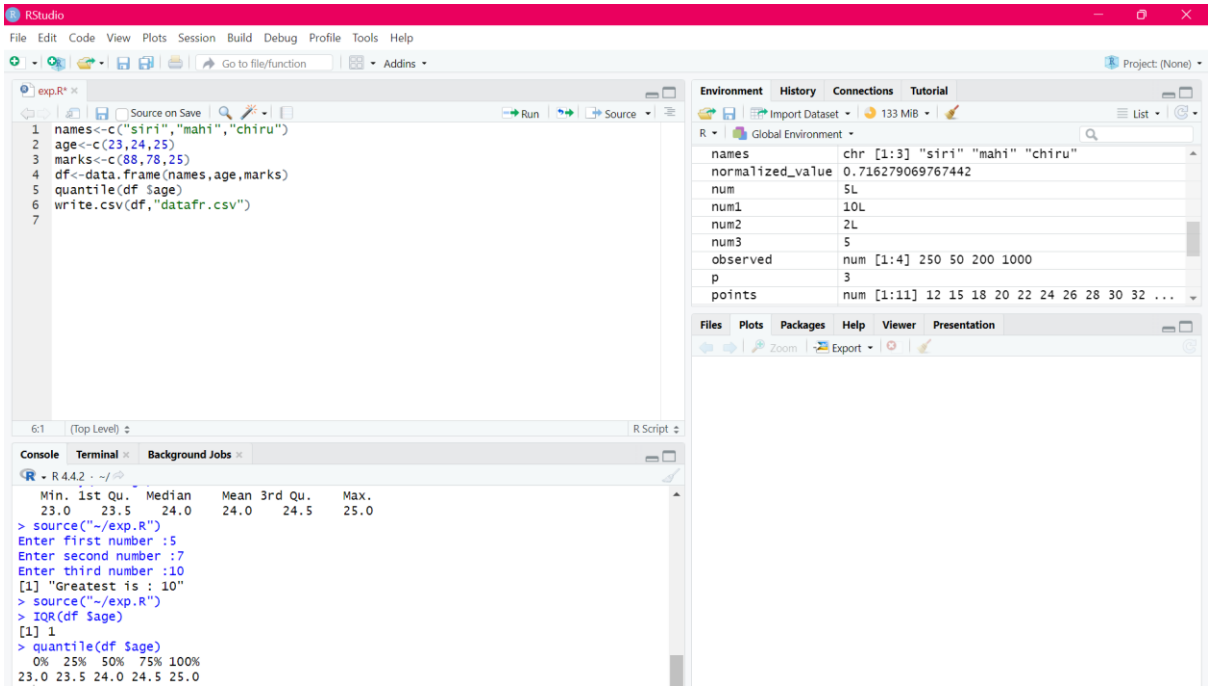
AIM:

To write the program for central tendency and data dispersion measures.

PROGRAM:

```
names<-c("siri","mahi","chiru")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
quantile(df $age)
write.csv(df,"datafr.csv")
```

OUTPUT:



RESULT:

Thus, the program for central tendency and data dispersion measures was executed successfully

EXPERIMENT-11

MID RANGE:

AIM:

To write the program for central tendency and data dispersion measures.

PROGRAM:

```
names <- c("siri", "mahi", "chiru")

age <- c(23, 24, 25)

marks <- c(88, 78, 25)

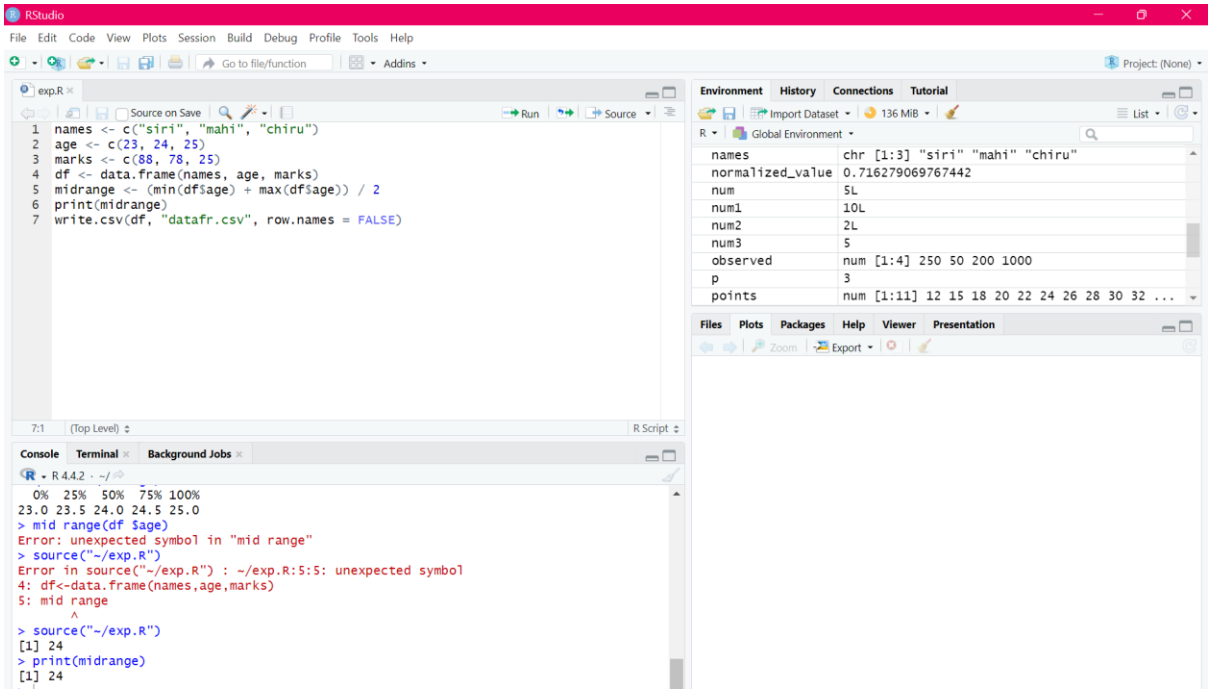
df <- data.frame(names, age, marks)

midrange <- (min(df$age) + max(df$age)) / 2

print(midrange)

write.csv(df,"datafr.csv")
```

OUTPUT:



RESULT:

Thus, the program for central tendency and data dispersion measures was executed successfully

EXPERIMENT-12

Z-SCOORE NORMALIZATION:

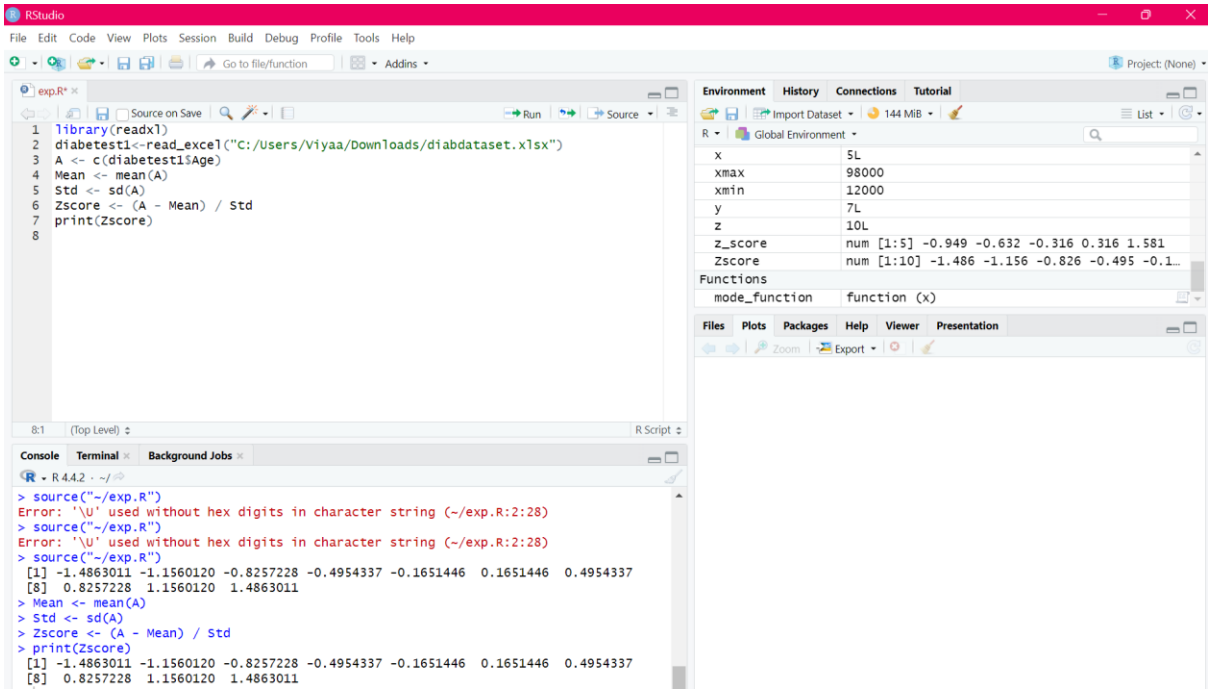
AIM:

To write the program for Z-scoore normalization using R-tool.

PROGRAM:

```
library(readxl)
diabetest1<-read_excel("C:/Users/Viyaa/Downloads/diabdataset.xlsx")
A <- c(diabetest1$Age)
Mean <- mean(A)
Std <- sd(A)
Zscore <- (A - Mean) / Std
print(Zscore)
```

OUTPUT:



RESULT:
Thus, the Z-score normalization using R tool was executed successfully.

EXPERIMENT-13

MIN,MAX,MEAN,MINMAX:
AIM:

To write the program for the minimum,maximum,mean and minmax using r-TOOL

PROGRAM:

MEAN

```
library(readxl)

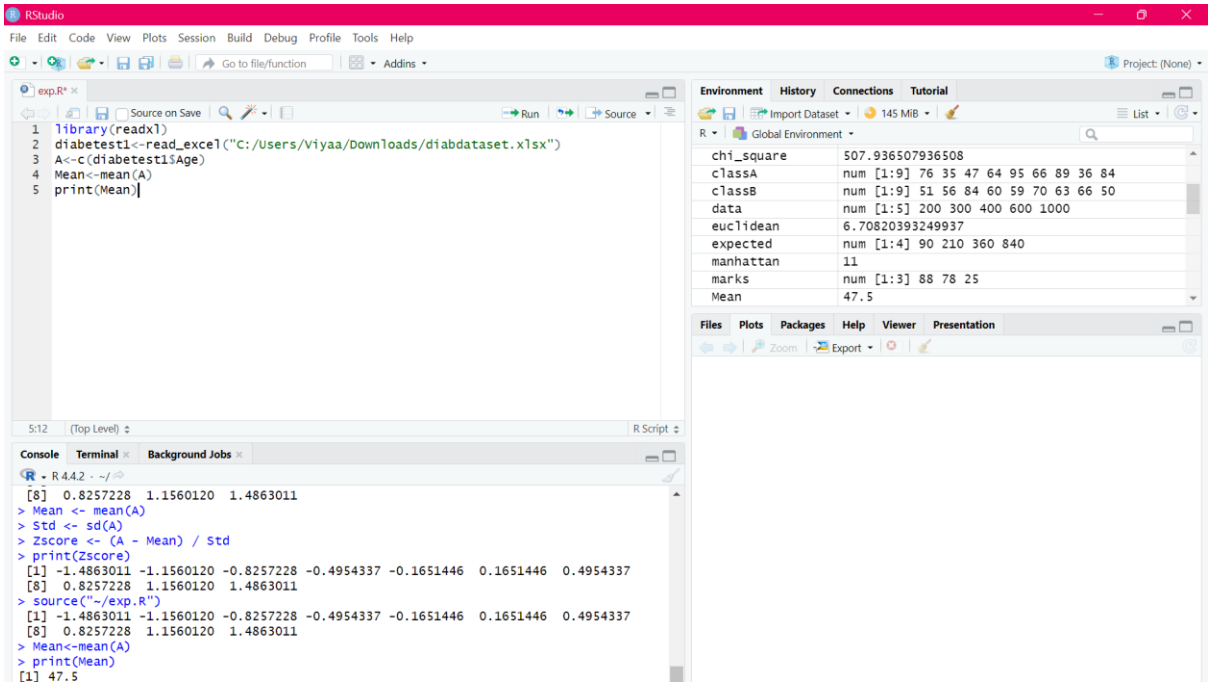
diabetest1<-read_excel("C:/Users/Viyaa/Downloads/diabdataset.xlsx")

A<-c(diabetest1$Age)

Mean<-mean(A)

print(Mean)
```

OUTPUT:



MINIMUM

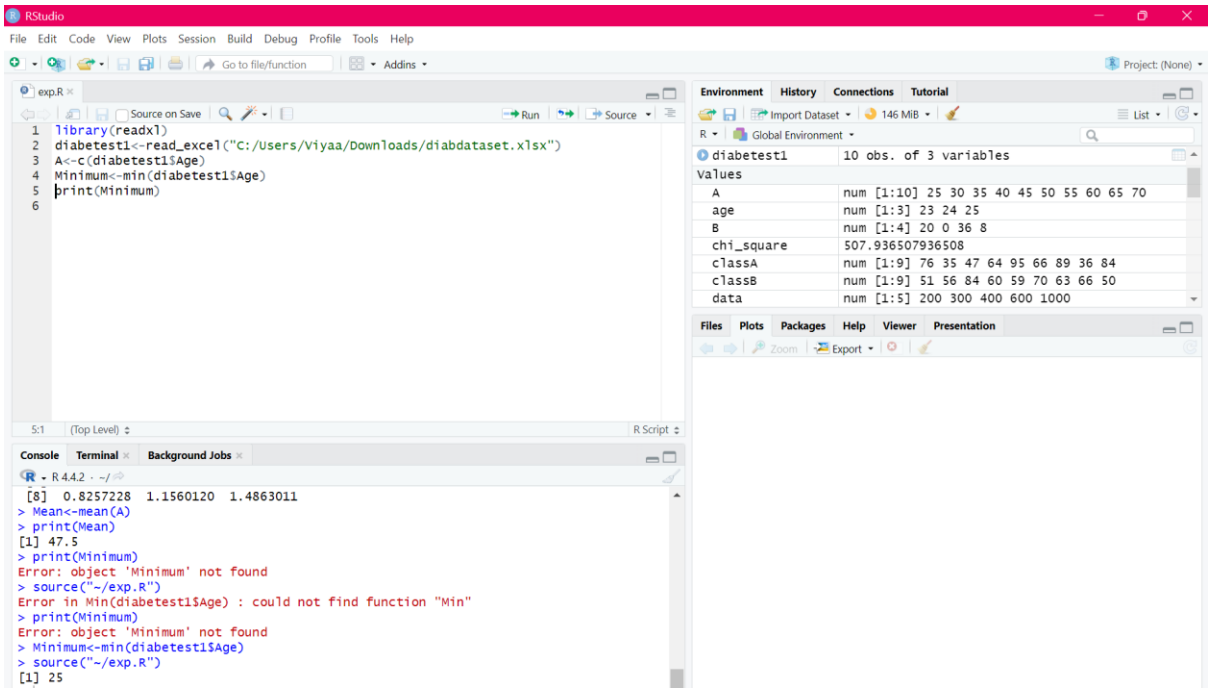
library(readxl)

diabetest1<-read_excel("C:/Users/Viyaa/Downloads/diabdataset.xlsx")

A<-c(diabetest1$Age)

Minimum<-min(diabetest1$Age)

print(Minimum)OUTPUT:



MAXIMUM

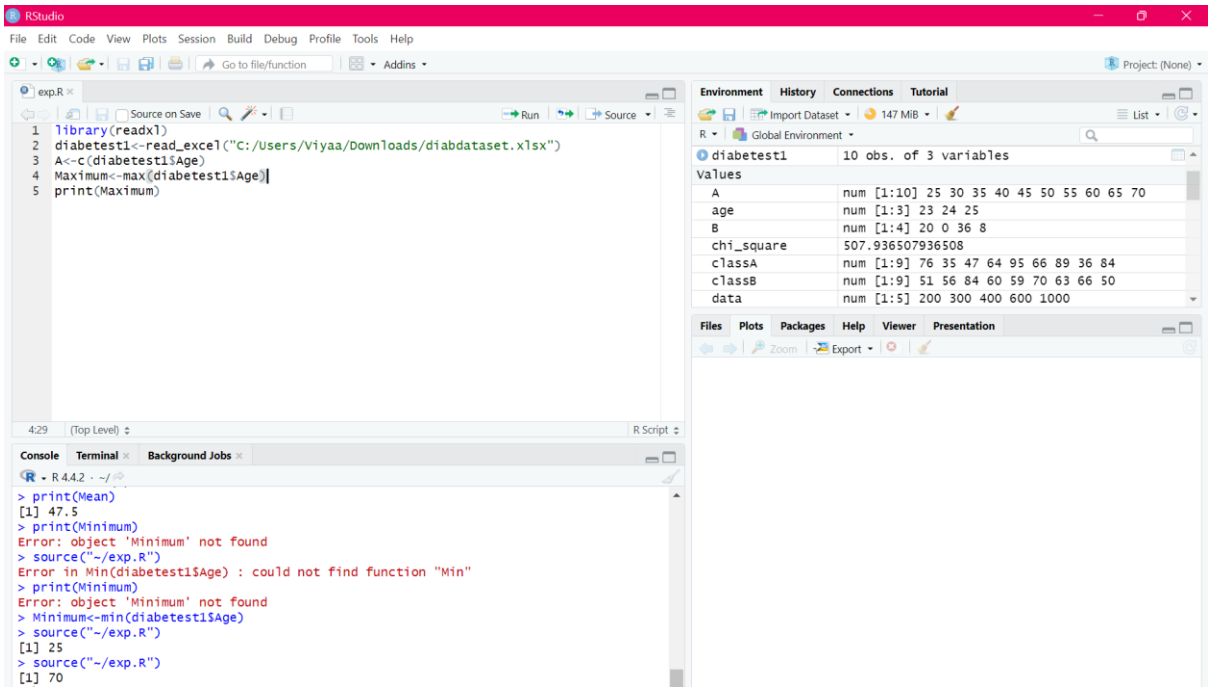
library(readxl)

diabetest1<-read_excel("C:/Users/Viyaa/Downloads/diabdataset.xlsx")

A<-c(diabetest1$Age)

Maximum<-max(diabetest1$Age)

print(Maximum)OUTPUT:



MINMAX

library(readxl)

diabetest1<-read_excel("C:/Users/Viyaa/Downloads/diabdataset.xlsx")

A<-c(diabetest1$Age)

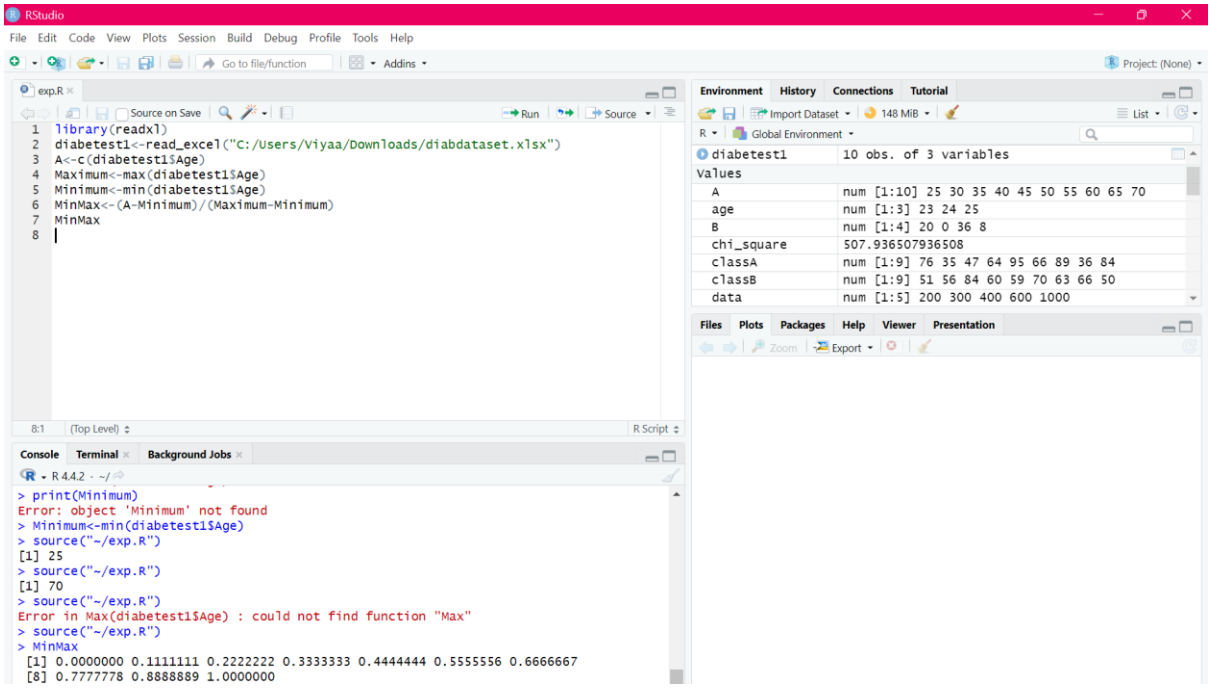
Maximum<-Max(diabetest1$Age)

Minimum<-Min(diabetest1$Age)

MinMax<-(A-Minimum)/(Maximum-Minimum)

MinMax

OUTPUT:



RESULT:

Thus, the program for min,max,minmax,mean was executed successfully.

EXPERIMENT-14

BAR PLOT AND HORIZONTAL BAR:

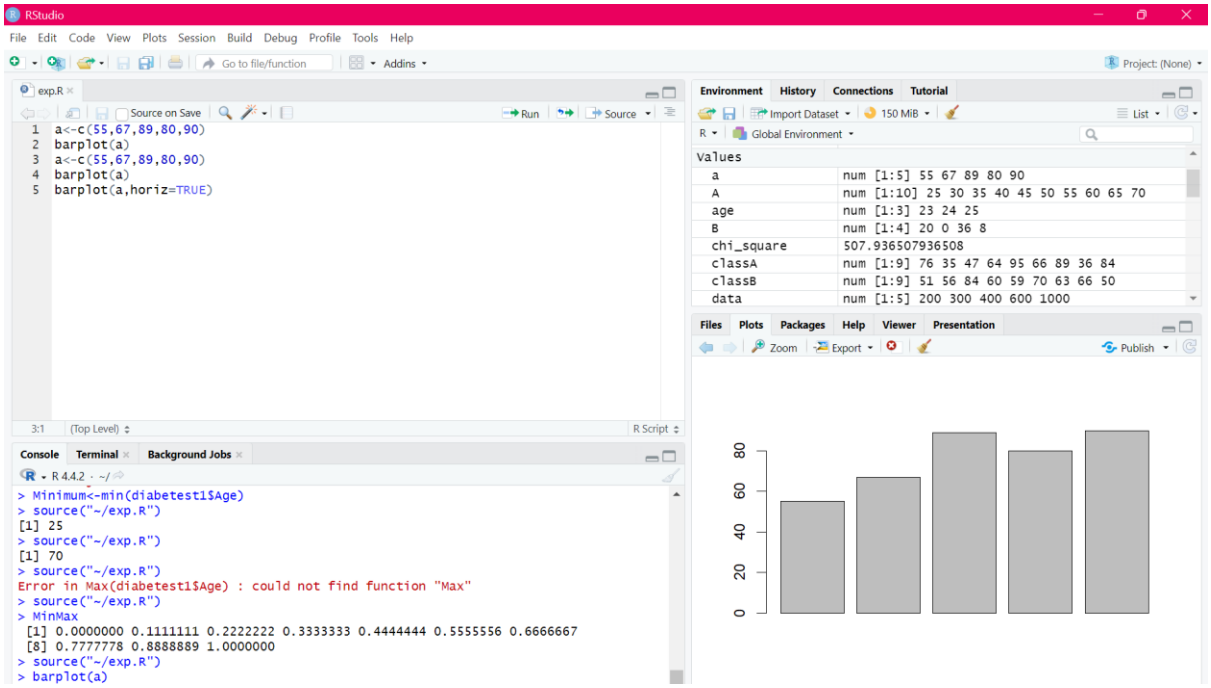
AIM:

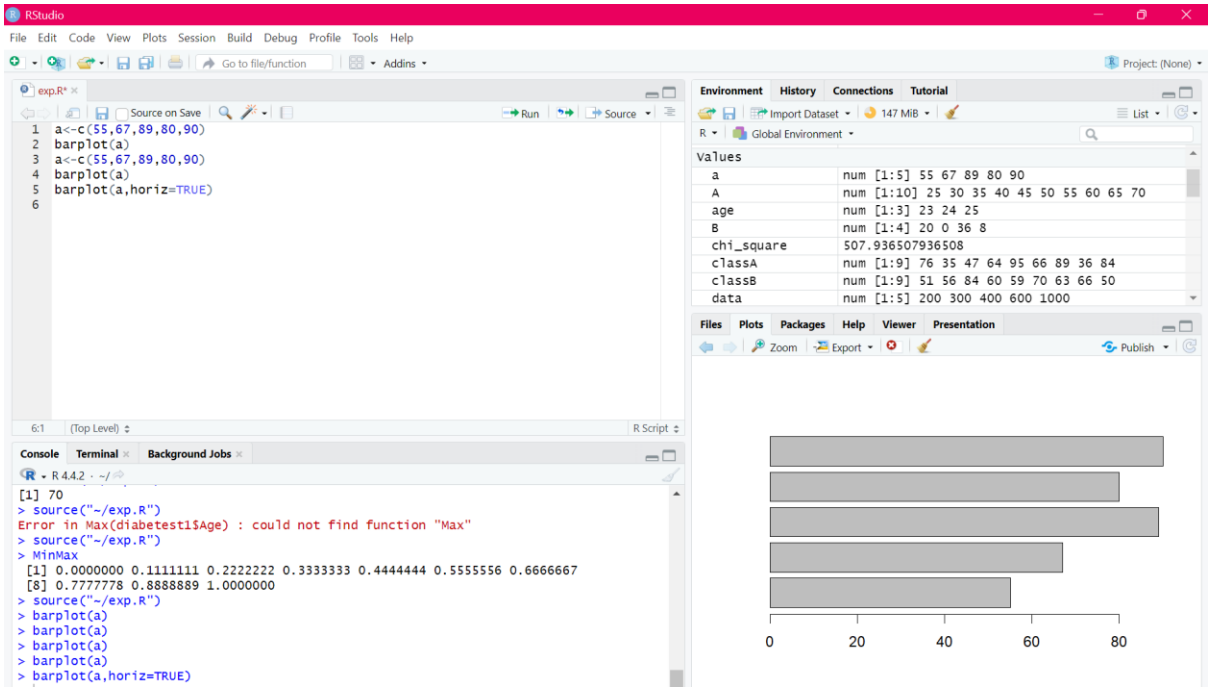
To draw the bar plot and horizontal bar using R-tool.

PROGRAM:

```
a<-c(55,67,89,80,90)
barplot(a)
a<-c(55,67,89,80,90)
barplot(a)
barplot(a,horiz=TRUE)
```

OUTPUT:





RESULT:

Thus, the bar and horizontal bar plot was executed successfully.

EXPERIMENT-15

BOX PLOT:

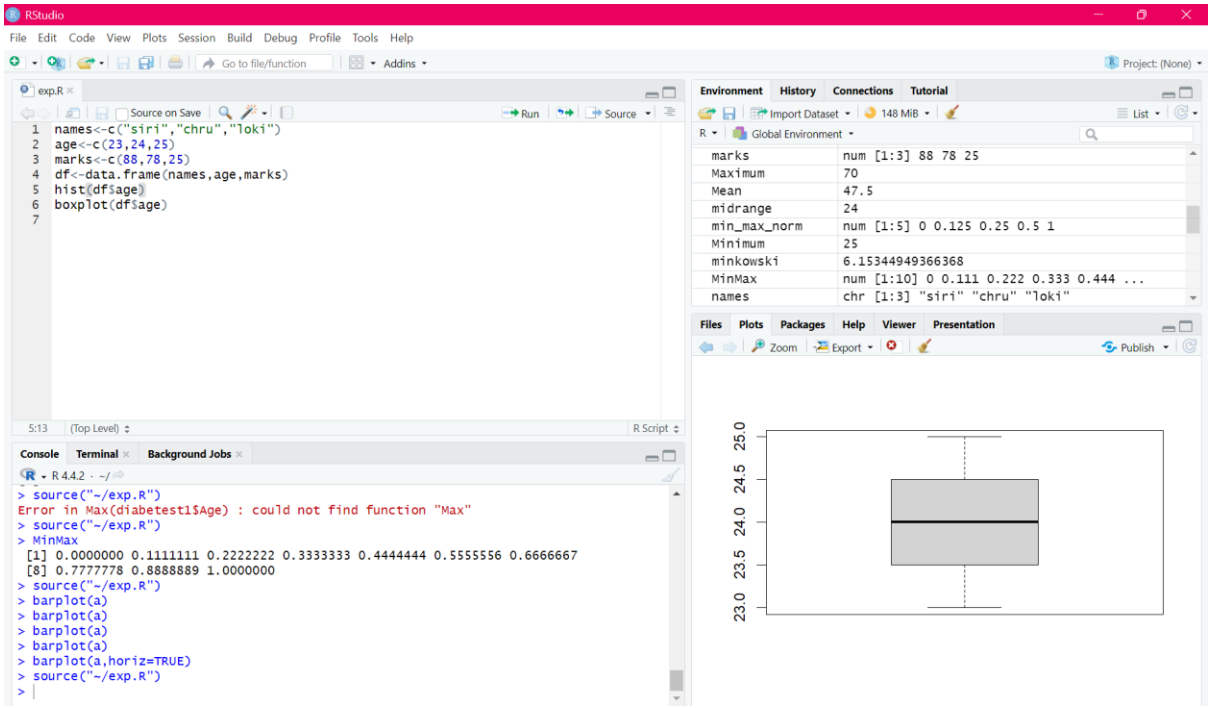
AIM:

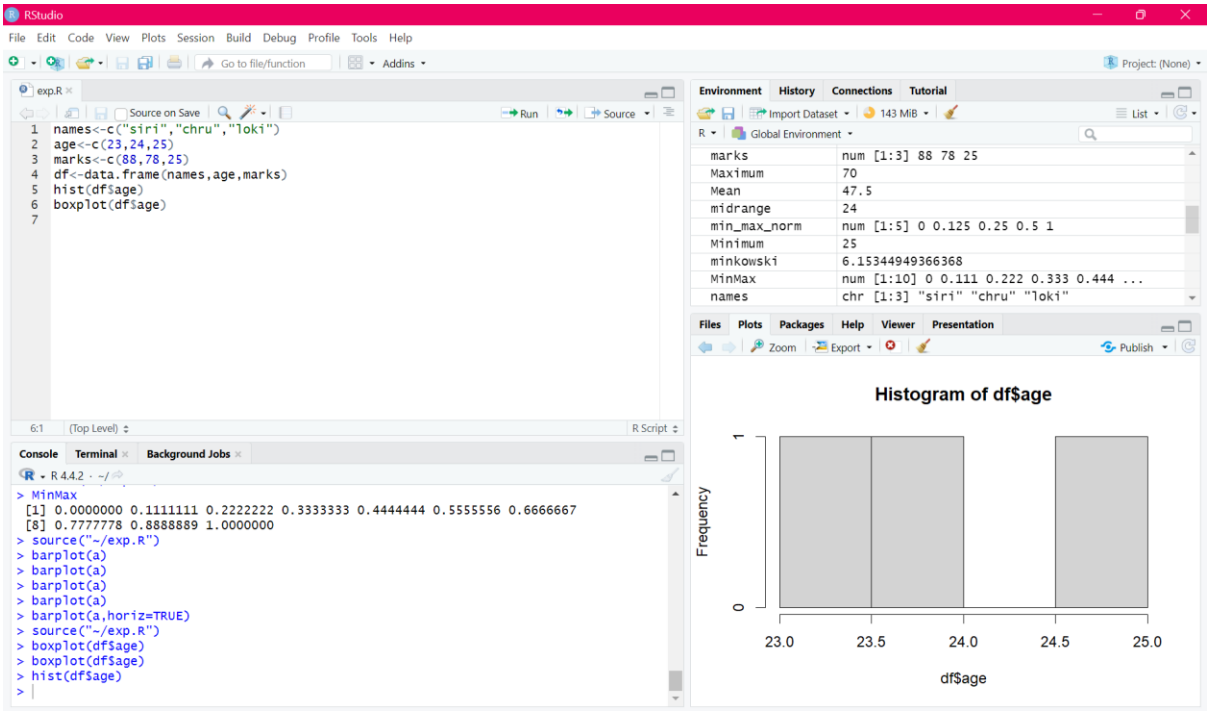
To draw the box plot using R-tool.

PROGRAM:

```
names<-c("siri","chru","loki")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
hist(df$age)
boxplot(df$age)
```

OUTPUT:





RESULT:

Thus, the box plot was executed successfully.

EXPERIMENT-16

HISTOGRAM:

AIM:

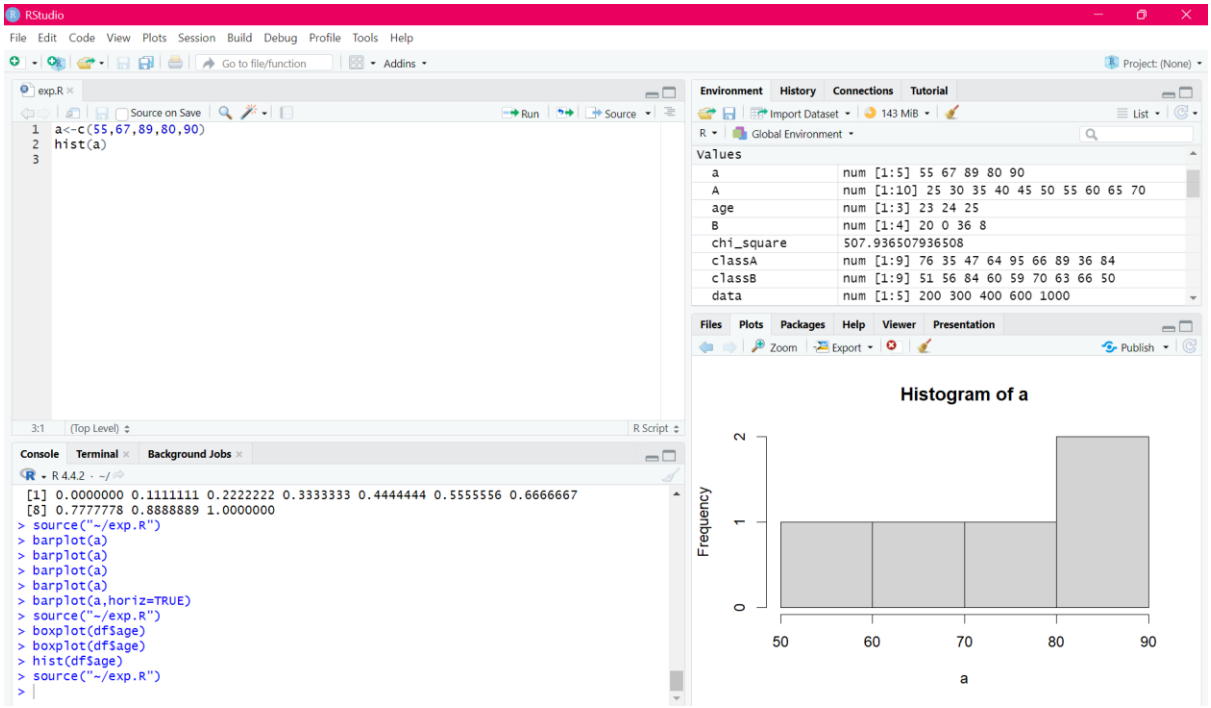
To draw the histogram plot using R-tooll.

PROGRAM:

a<-c(55,67,89,80,90)

hist(a)

OUTPUT:



RESULT:

Thus, the histogram plot was executed successfully.

EXPERIMENT-17

CORRELATION ANALYSIS:

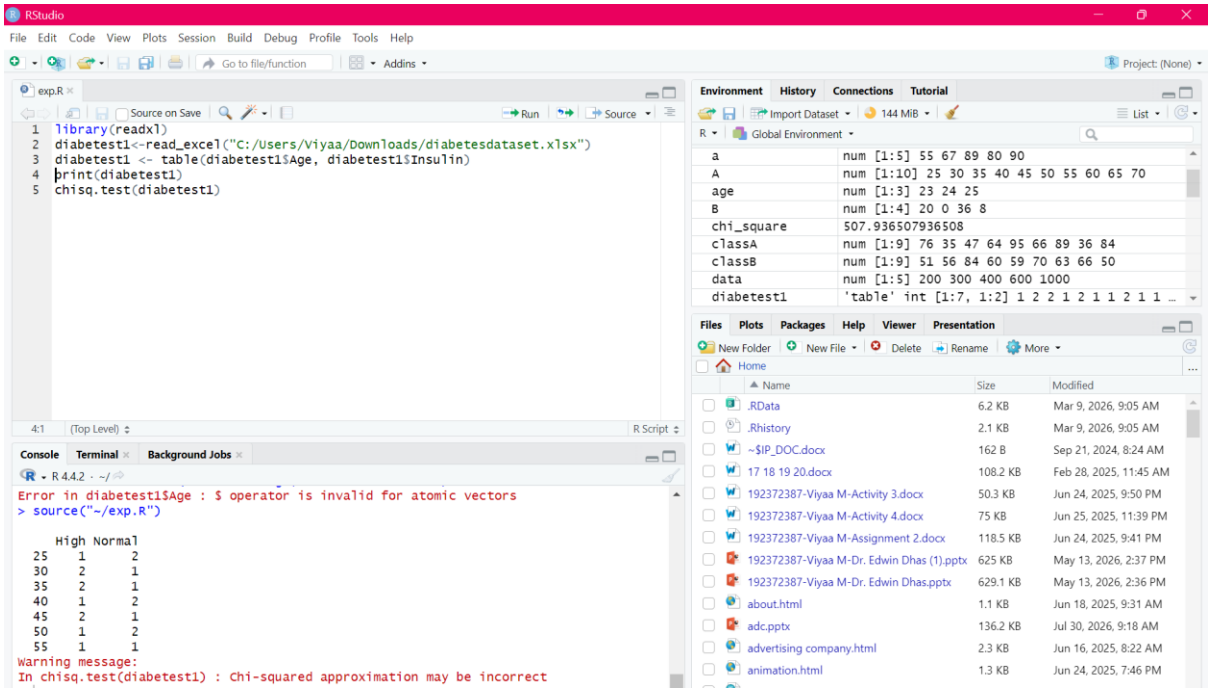
AIM:

To write the program for correlation analysis using R-tool.

PROGRAM:

```
library(readxl)
diabetest1<-read_excel("C:/Users/Viyaa/Downloads/diabetesdataset.xlsx")
diabetest1 <- table(diabetest1$Age, diabetest1$Insulin)
print(diabetest1)
chisq.test(diabetest1)
```

OUTPUT:



RESULT:

Thus, the correlation analysis was executed successfully.

EXPERIMENT-18

SCATTER PLOT:

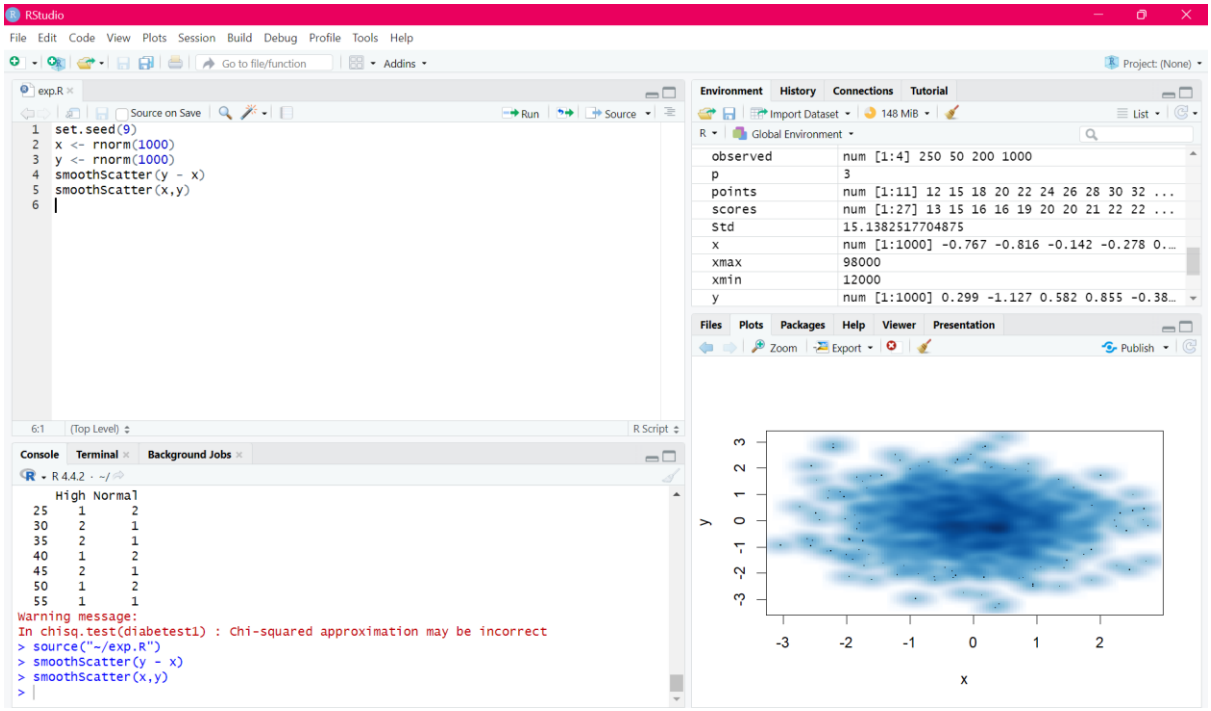
AIM:

To draw the scatter plot using R-tool

PROGRAM:

```
set.seed(9)
x <- rnorm(1000)
y <- rnorm(1000)
smoothScatter(y - x)
smoothScatter(x,y)
```

OUTPUT:



RESULT:

Thus, the scatter plot was executed successfully.

EXPERIMENT-19

LINEAR REGRESSION:

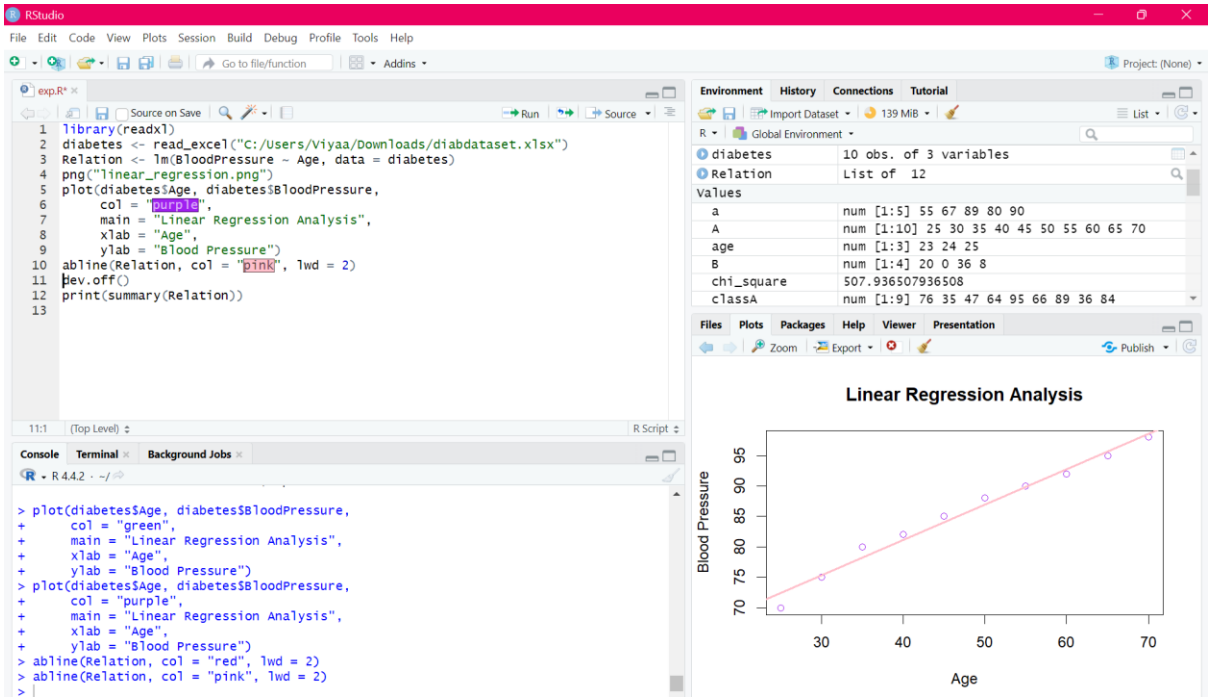
AIM:

To write thr program for the linear regression using R-tool.

PROGRAM:

```
library(readxl)
diabetes <- read_excel("C:/Users/Viyaa/Downloads/diabdataset.xlsx")
Relation <- lm(BloodPressure ~ Age, data = diabetes)
png("linear_regression.png")
plot(diabetes$Age, diabetes$BloodPressure,
    col = "purple",
    main = "Linear Regression Analysis",
    xlab = "Age",
    ylab = "Blood Pressure")
abline(Relation, col = "pink", lwd = 2)
dev.off()
```

print(summary(Relation))OUTPUT:



RESULT:

Thus, the linear regression program was executed successfully.

EXPERIMENT-20

MULTIPLE REGRESSION:

AIM:

To write the program for the multiple regression.

PROGRAM:

```
library(readxl)
diabetes <- read_excel("C:/Users/Viyaa/Downloads/diabdataset.xlsx")
input <- diabetes[, c("Age", "BloodPressure", "Glucose")]
model <- lm(Age ~ BloodPressure + Glucose, data = input)
print(model)
A <- coef(model)[1]
print(A)
xBloodPressure <- coef(model)[2]
yGlucose <- coef(model)[3]
print(xBloodPressure)
print(yGlucose)
```

```
y <- paste("Age =", A, "+", xBloodPressure,
           "* BloodPressure +", yGlucose, "* Glucose")
print(y)
```

OUTPUT:

